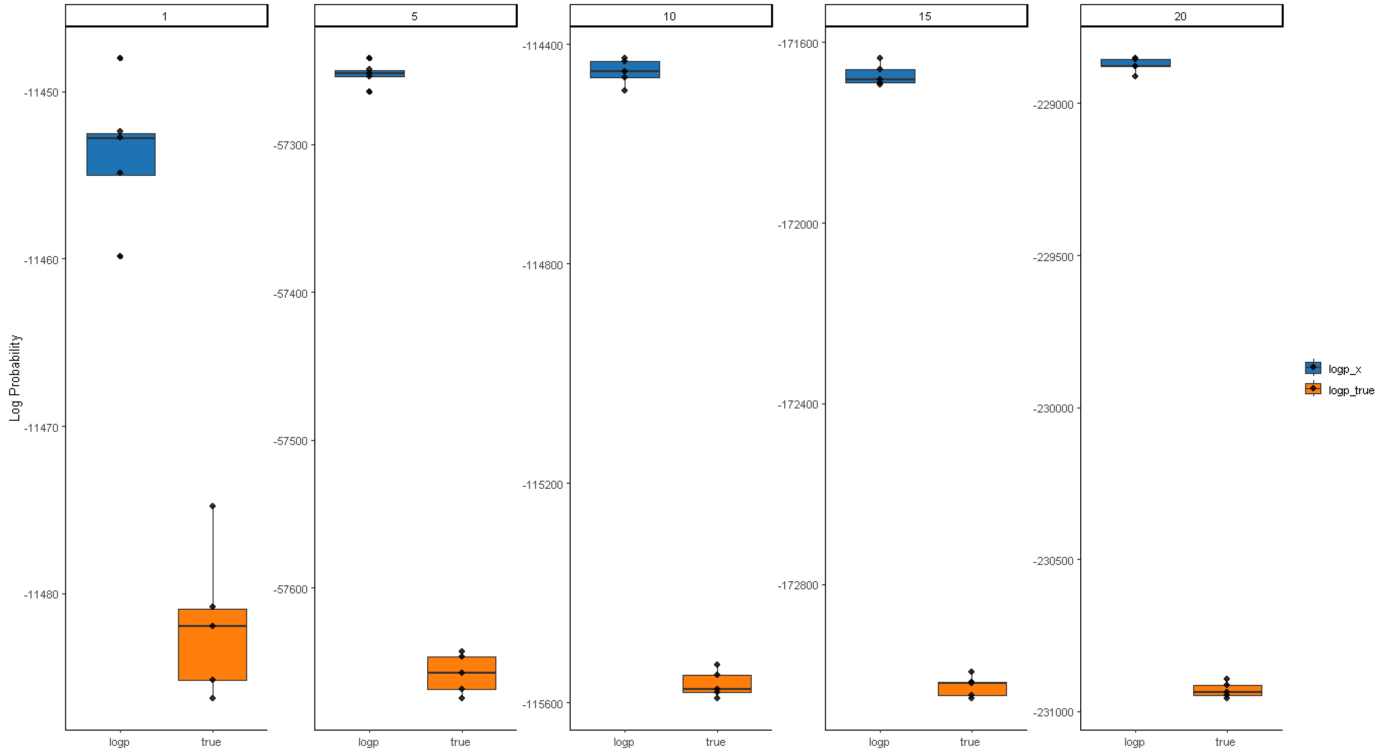
20260818

Benchmarking gLike

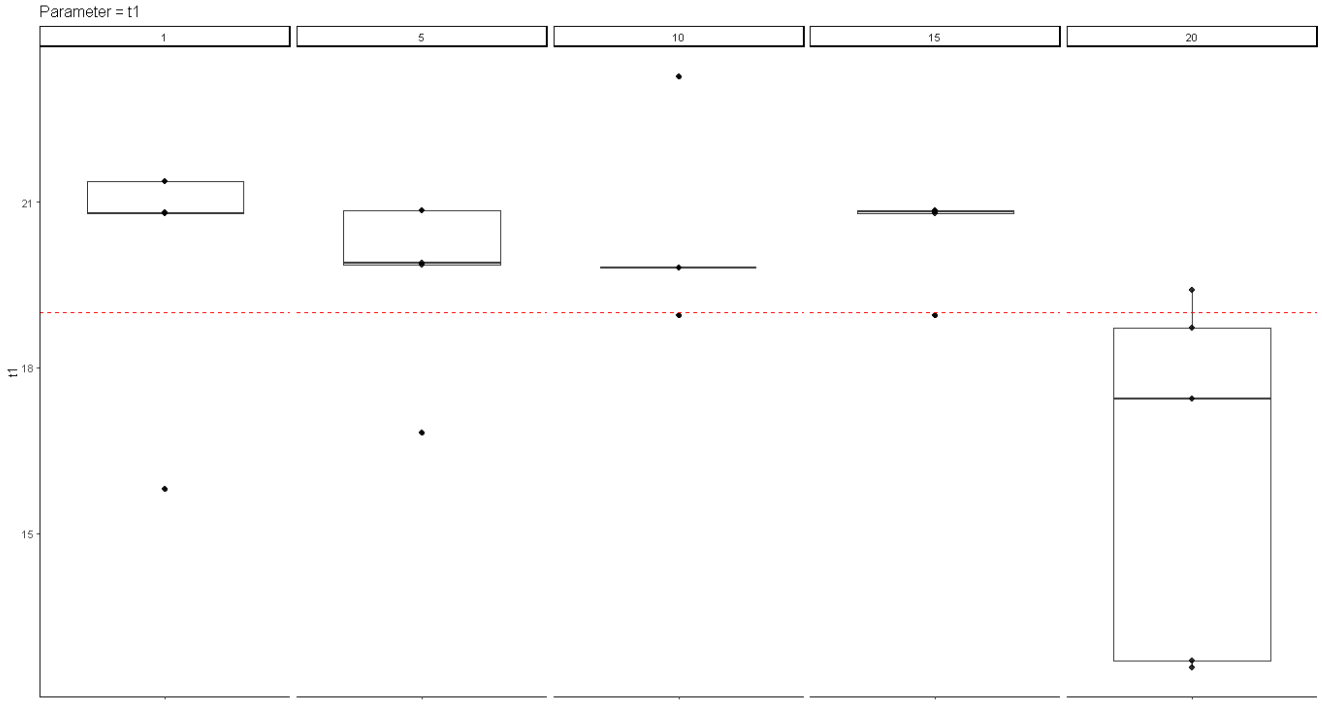
- I used prior set-up in which the starting initial conditions where the true simulation parameters.
- I then ran 5 replicates with 1, 5, 10, 15, and 20 equidistant trees (the replicate number is equivalent to the same simulation)
- I scraped the *.log output to form a `.csv` with the log likelihoods:
  - glike_optimize > opt_results_glike_true_rep > benchmarks.csv > data

```
file,NUM_TREES,runtime,logp,true
ntree_10_rep_1_simulate_ARG.log,10,4h 10m 44.3s,-114450.39086527425,-115575.30724909506
ntree_10_rep_2_simulate_ARG.log,10,2h 50m 44.7s,-114430.89589196378,-115549.6971735544
ntree_10_rep_3_simulate_ARG.log,10,3h 11m 57.9s,-114460.98366228075,-115591.60542488196
ntree_10_rep_4_simulate_ARG.log,10,4h 2m 37.8s,-114425.49448688763,-115580.96402784876
ntree_10_rep_5_simulate_ARG.log,10,4h 5m 47.1s,-114484.63318187867,-115531.44158329508
ntree_15_rep_1_simulate_ARG.log,15,3h 32m 24.1s,-171682.28940345277,-172993.05745085128
ntree_15_rep_2_simulate_ARG.log,15,3h 20m 19.7s,-171694.06636180426,-173017.01741842204
ntree_15_rep_3_simulate_ARG.log,15,3h 26m 53.5s,-171689.2326577323,-173051.14220182932
ntree_15_rep_4_simulate_ARG.log,15,3h 38m 22.5s,-171661.10573800188,-173045.946422306
ntree_15_rep_5_simulate_ARG.log,15,3h 55m 38.4s,-171635.76353715133,-173018.04837687744
ntree_1_rep_1_simulate_ARG.log,1,1h 14m 58.7s,-11452.473967799393,-11486.13318118242
ntree_1_rep_2_simulate_ARG.log,1,1h 56m 26.6s,-11452.8241253319,-11481.970983148289
ntree_1_rep_3_simulate_ARG.log,1,1h 40m 27.2s,-11448.036547576856,-11485.169600668914
ntree_1_rep_4_simulate_ARG.log,1,1h 44m 30.9s,-11454.965250230922,-11474.778208985264
ntree_1_rep_5_simulate_ARG.log,1,1h 3m 22.3s,-11459.82610059604,-11480.900415908938
ntree_20_rep_1_simulate_ARG.log,20,3h 9m 39.5s,-228877.01093107346,-230937.8534065606
ntree_20_rep_2_simulate_ARG.log,20,3h 24m 29.8s,-228879.09984673653,-230955.92361566465
ntree_20_rep_3_simulate_ARG.log,20,2h 21m 0.0s,-228912.7799594752,-230946.37433518542
ntree_20_rep_4_simulate_ARG.log,20,5h 9m 20.8s,-228853.63183211716,-230913.5433979332
ntree_20_rep_5_simulate_ARG.log,20,4h 47m 4.1s,-228851.54946415598,-230892.58785488174
ntree_5_rep_1_simulate_ARG.log,5,3h 12m 16.6s,-57264.695542326626,-57674.39884012138
ntree_5_rep_2_simulate_ARG.log,5,3h 15m 8.4s,-57253.79518269624,-57643.32924301359
ntree_5_rep_3_simulate_ARG.log,5,3h 0m 13.5s,-57241.68107011021,-57657.846733991915
ntree_5_rep_4_simulate_ARG.log,5,3h 18m 30.7s,-57252.29544184288,-57646.933873911395
ntree_5_rep_5_simulate_ARG.log,5,2h 49m 47.4s,-57249.67893264933,-57668.843855334526
```



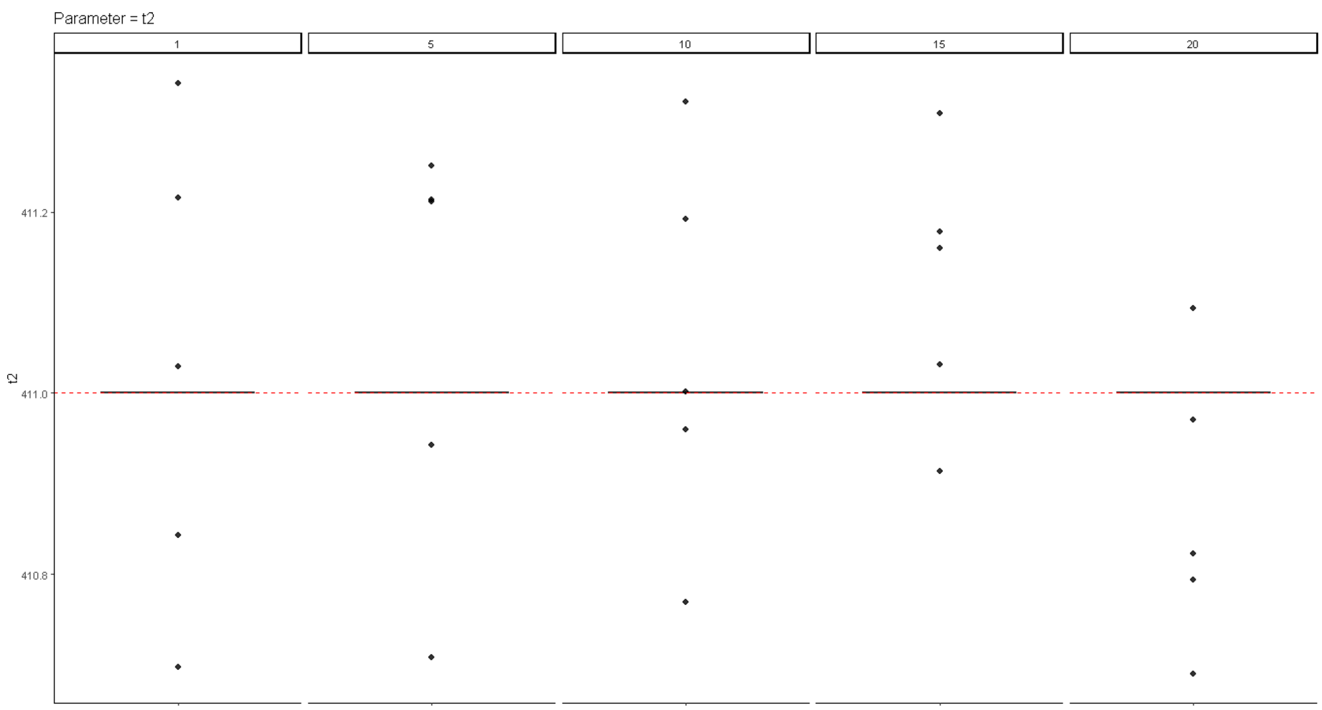
-
  - A priori, we found that there exist cases in which the best-fit ARG parameters yield a better log likelihood than the true underlying simulation parameters
  - We sought to further investigate if this is a consistent trend by looking through a different number of equidistant trees for the ARG --> the intuition is that
  - Figure: Each vertical facet represents a different number of `NUM_TREES`
    - Blue data represents the `logp_x` of the best-fit params
    - Orange data represents the `logp_true` of the simulation
- I scraped the *.log output to form a `.csv` with the parameters and compared the output to the true parameters
  - Figures: Each vertical facet represents a different number of `NUM_TREES`
    - The dashed red line indicates the true underlying simulation parameter

○



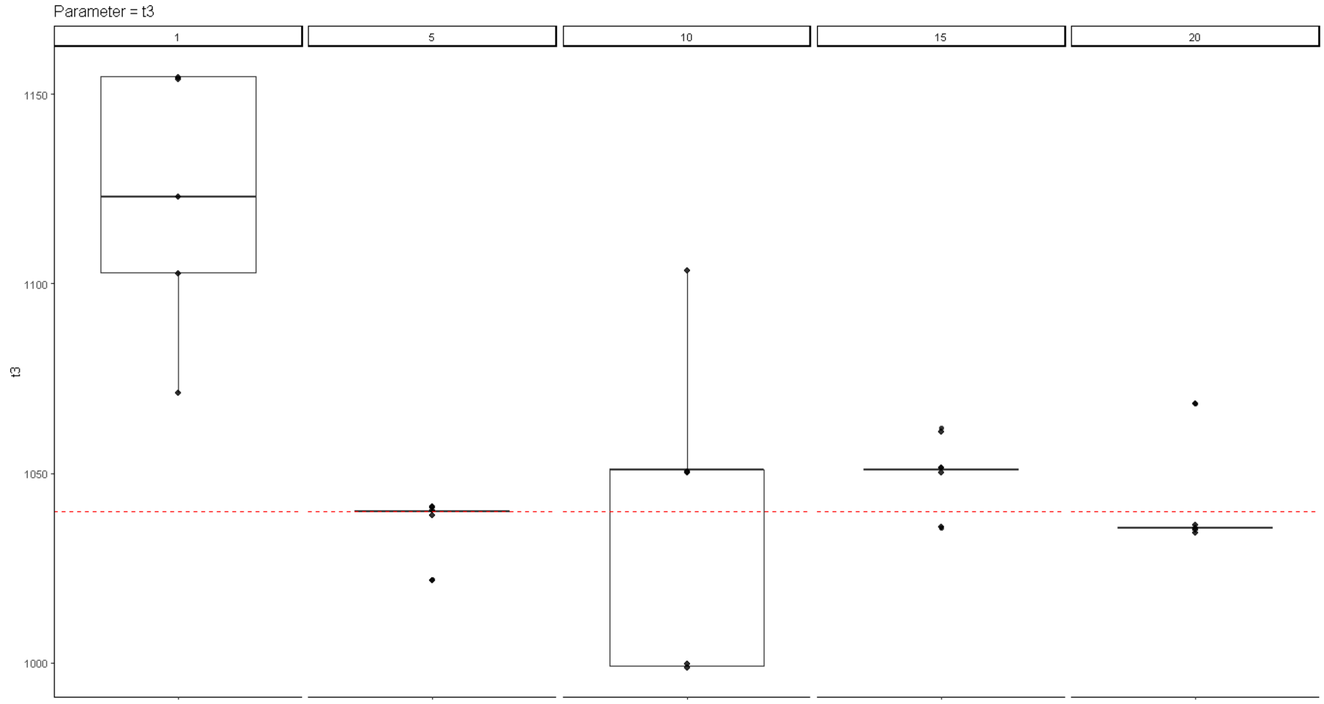
Parameter = t1
1
5
10
15
20
t1
21
18
15

○

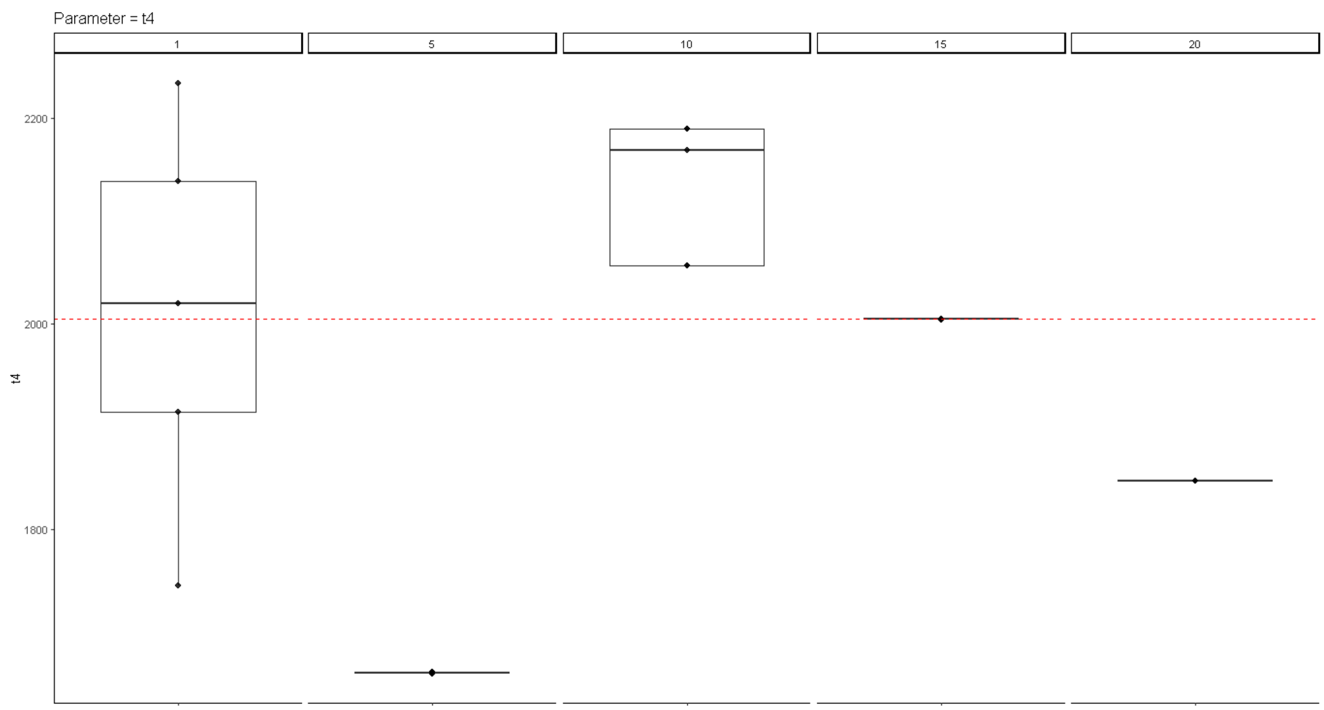


Parameter = t2
1
5
10
15
20
t2
411.2
411.0
410.8

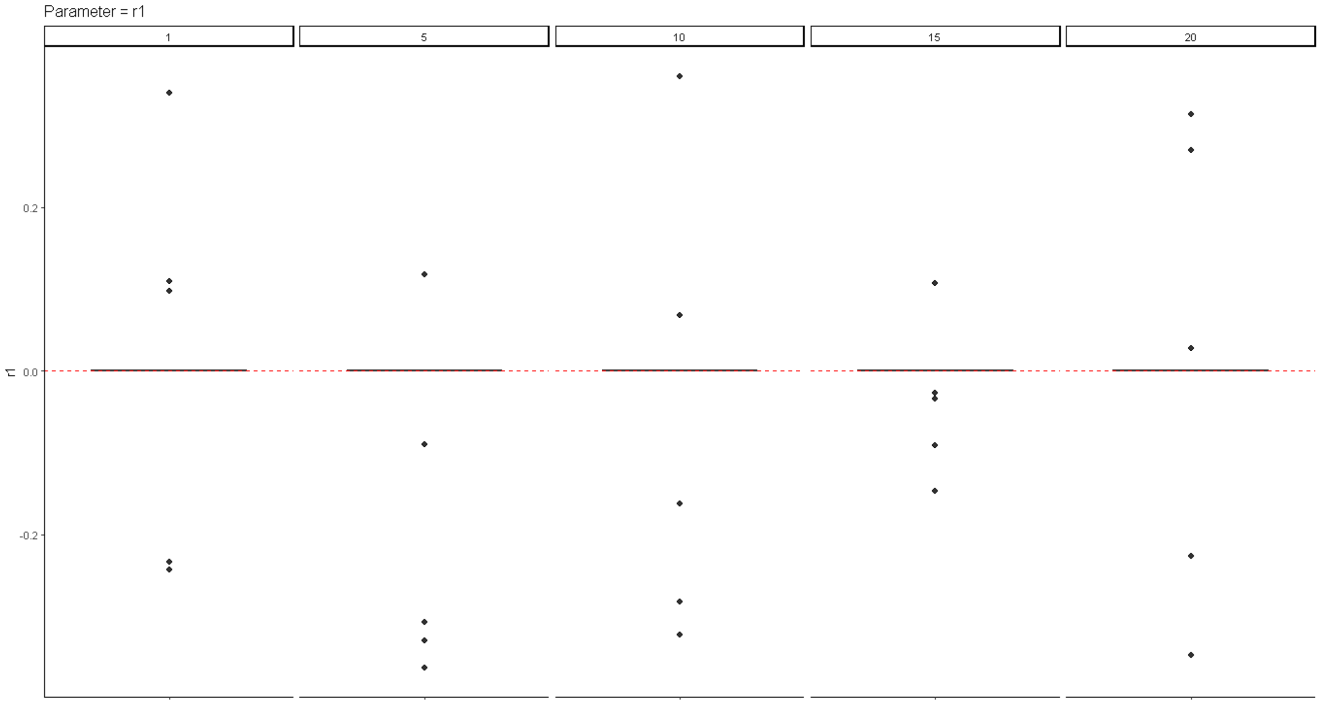
○


Parameter = t3
1
5
10
15
20
1150
1100
1050
1000
t3

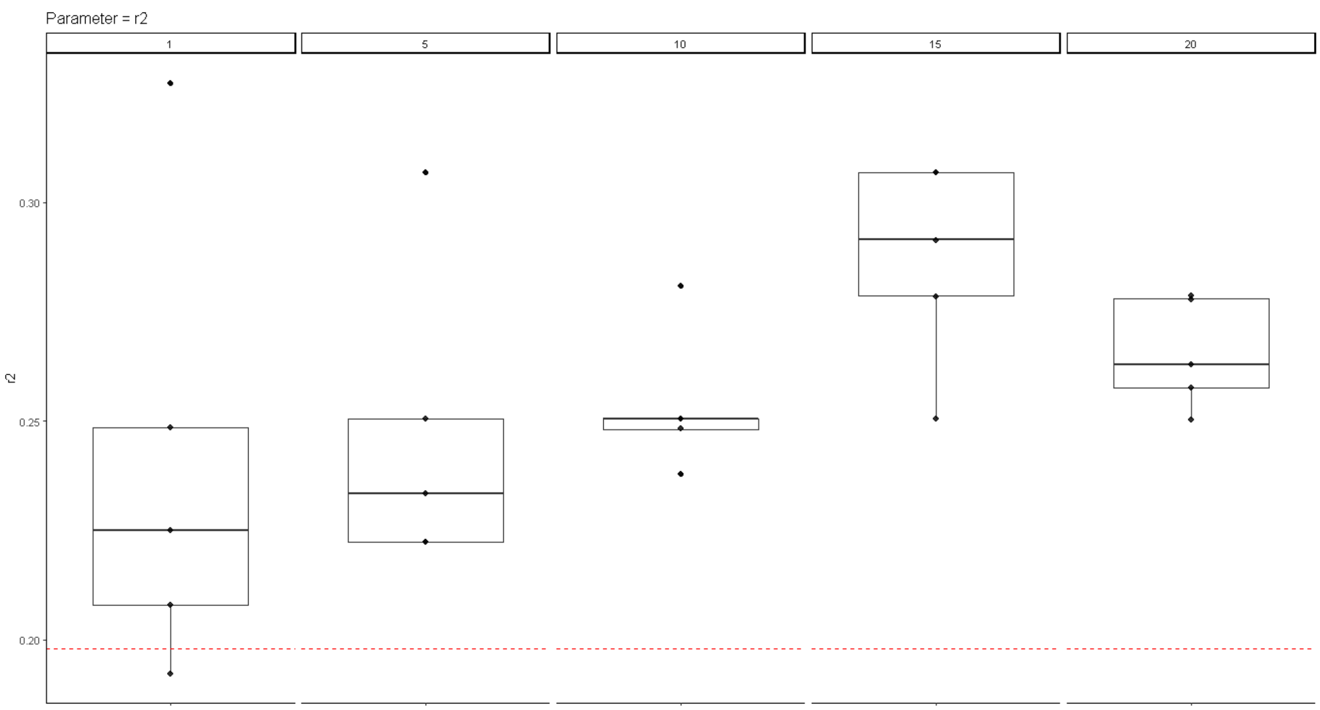
○


Parameter = t4
1
5
10
15
20
2200
2000
1800
t4

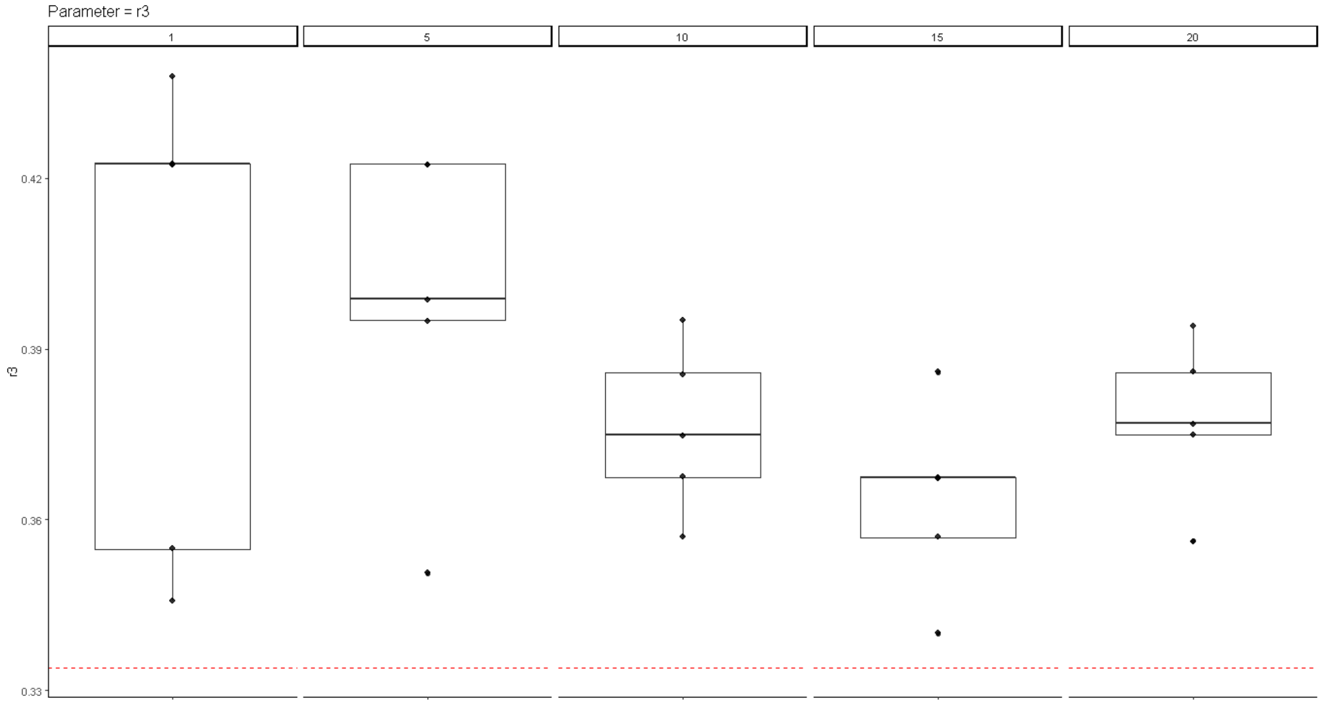
○


Parameter = r1
1
5
10
15
20
0.2
0.0
-0.2
r1

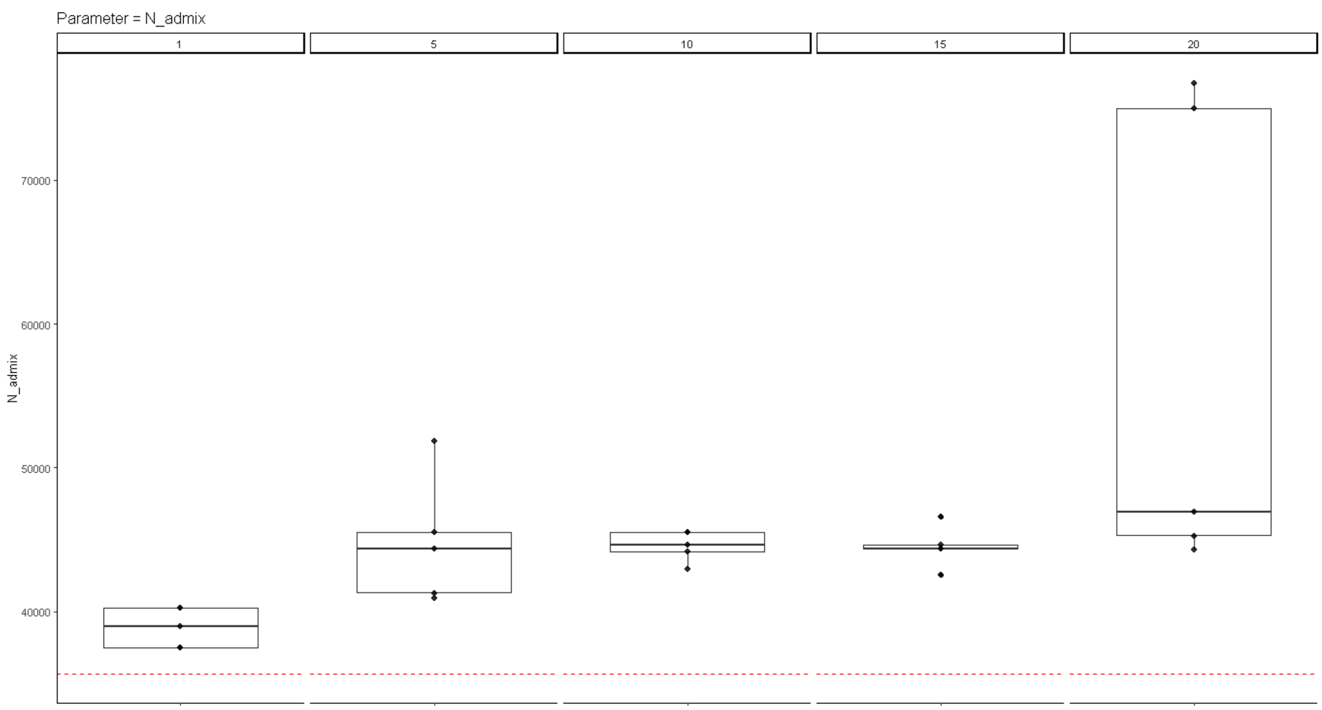
○


Parameter = r2
1
5
10
15
20
0.30
0.25
0.20
r2

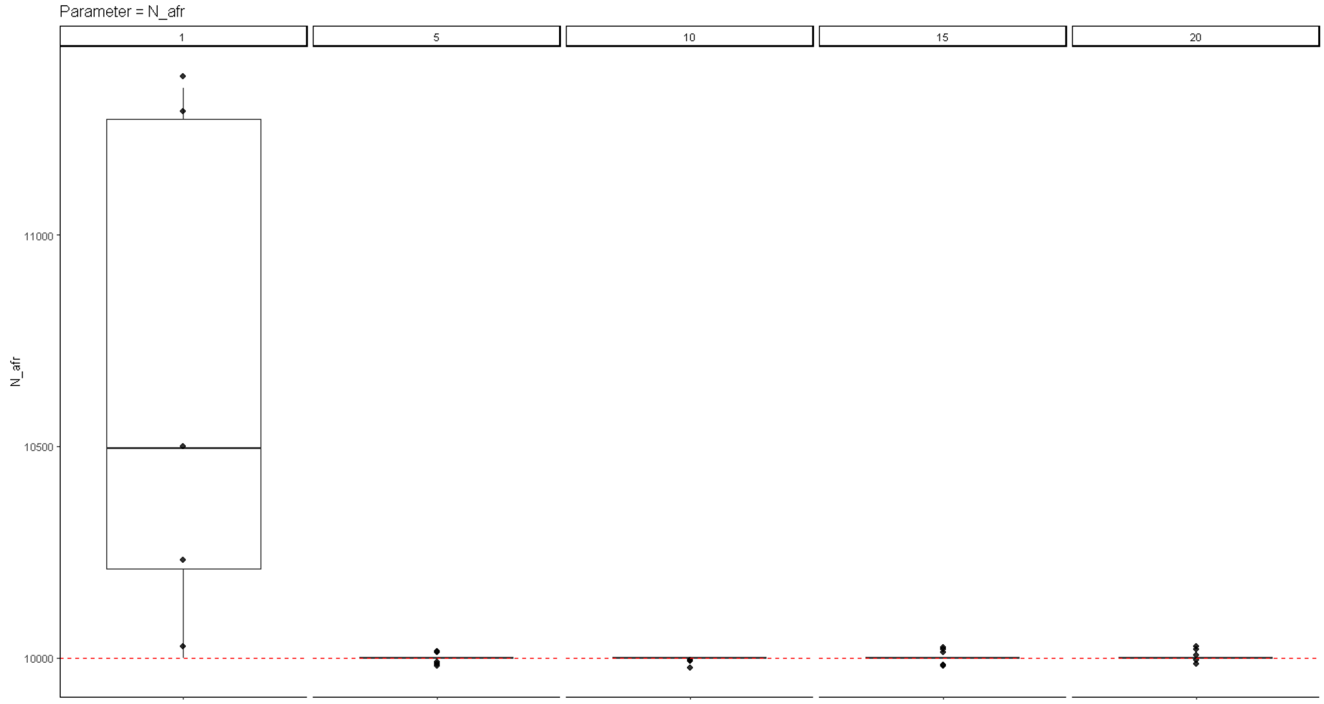
o


Parameter = r3
1
5
10
15
20
r3
0.42
0.39
0.36
0.33

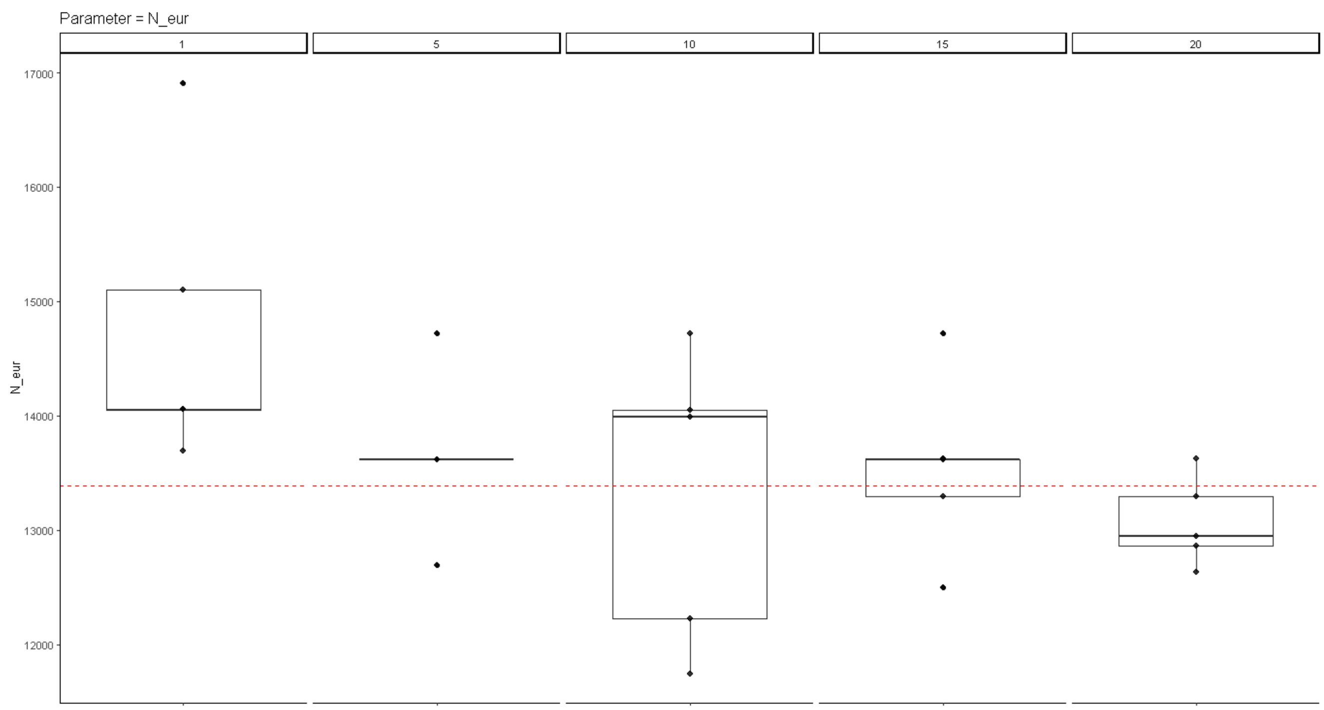
o


Parameter = N_admix
1
5
10
15
20
N_admix
70000
60000
50000
40000

○

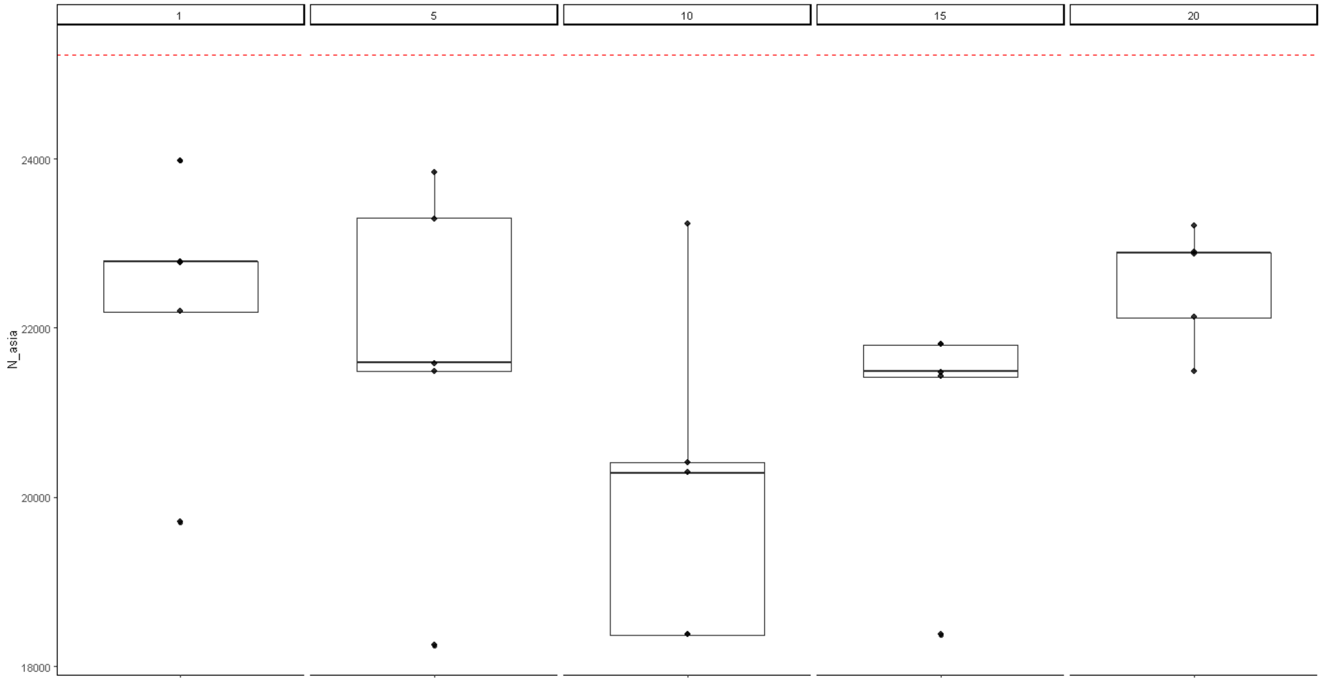

Parameter = N_afr
1
5
10
15
20
N_afr
11000
10500
10000

○


Parameter = N_eur
1
5
10
15
20
N_eur
17000
16000
15000
14000
13000
12000

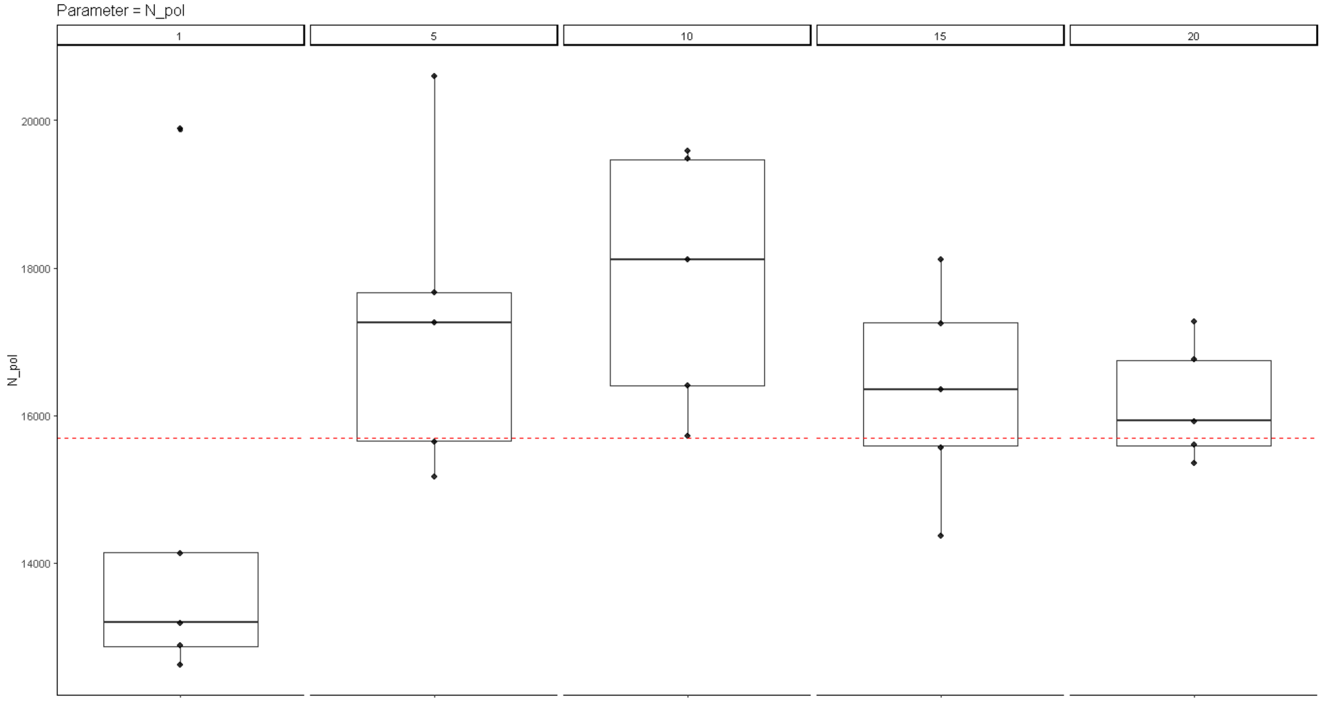
o

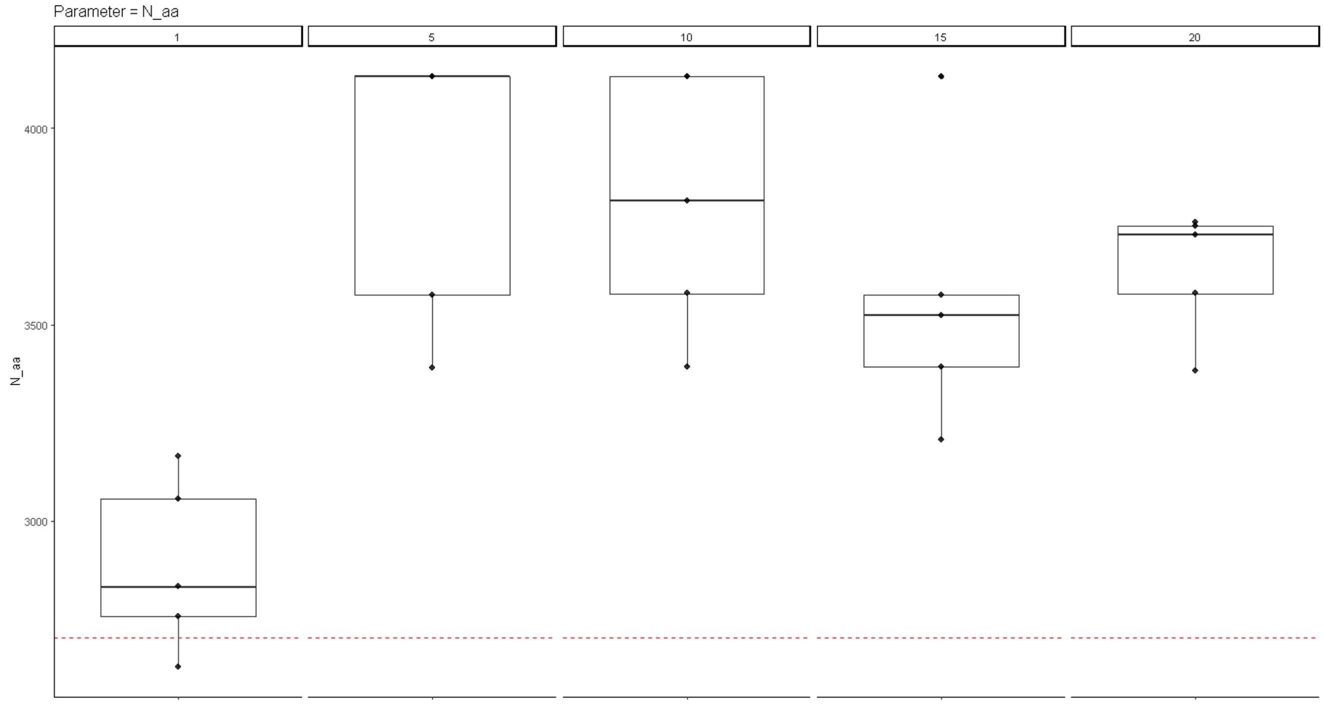
Parameter = N_asia



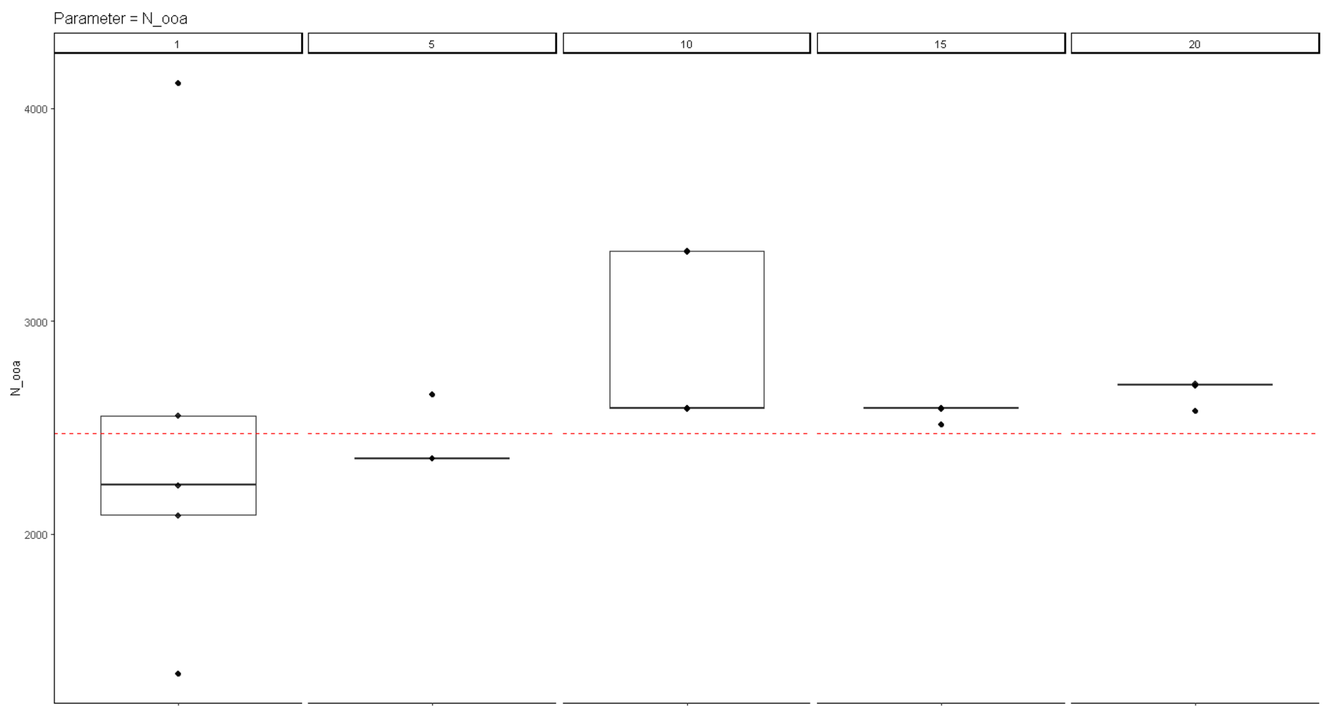
o

Parameter = N_pol

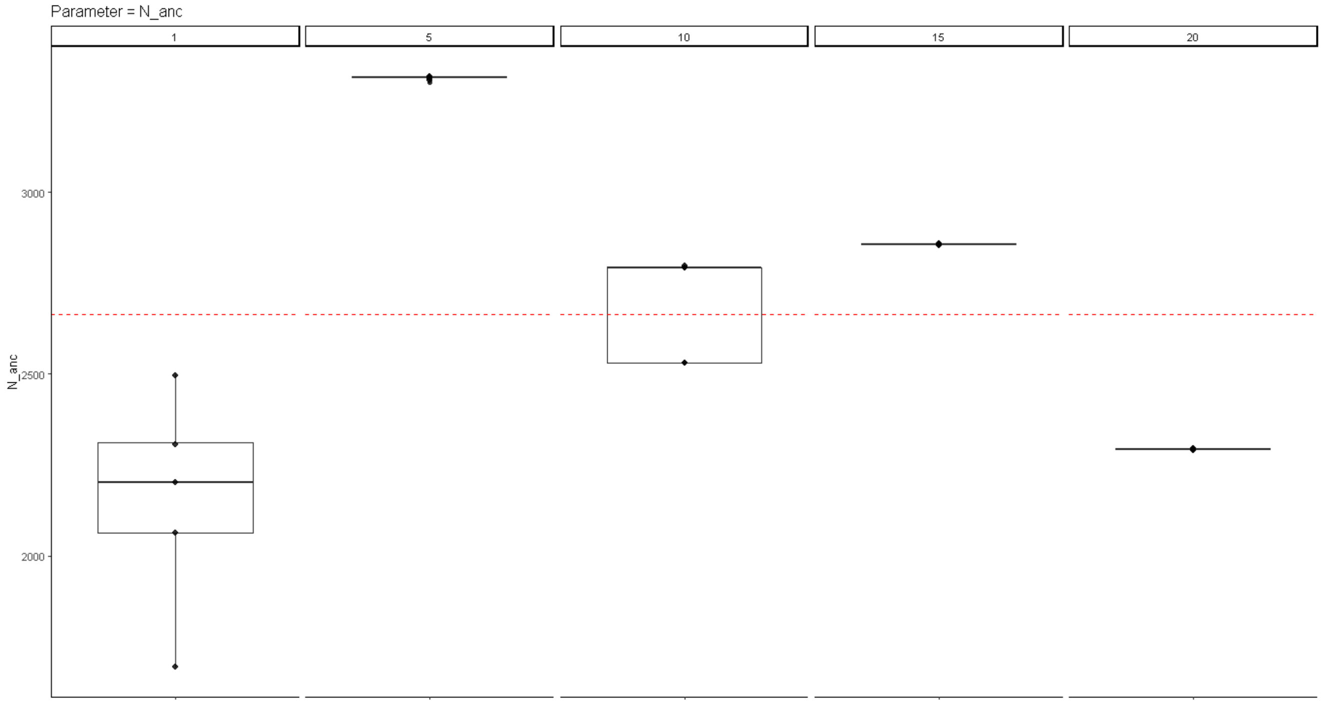


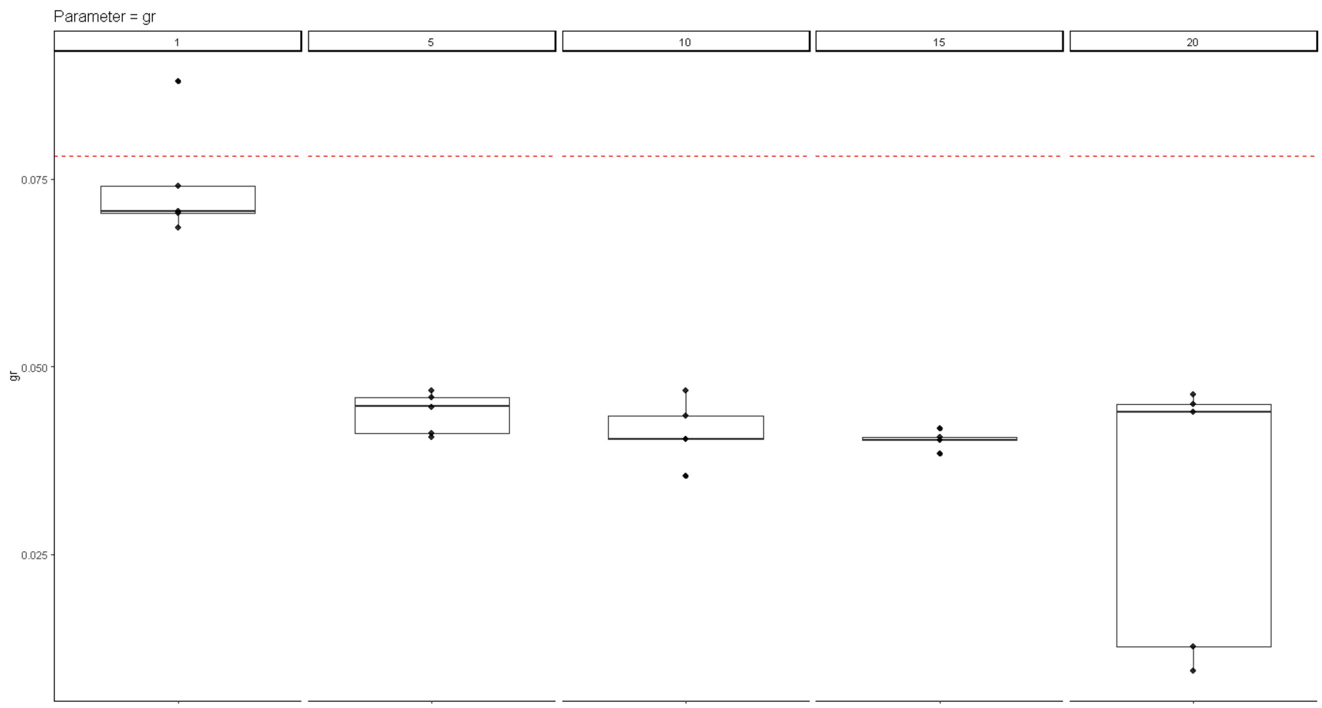
- 

Parameter = N_aa
1
5
10
15
20
4000
3500
3000
N_aa

- 

Parameter = N_ooa
1
5
10
15
20
4000
3000
2000
N_ooa

- 

- 

- TODO
  - Boxplots with percentage error or percentage bias
    - Arbitrary initial conditions, what is the relative error?
    - True initial conditions, what is the relative error?
  - Boxplots for loglikelihood comparison for both arbitrary and true initial starting conditions
  - Which model should we be using? Complicated or simple?
    - 3G09 three-population model without admixture
  - Finish / fix (?) CMA-ES -- Do we want to test this on which model?
    - To address param coupling
  - Fix r1, r2, r3 boundaries

- Look into Kappa (number of sampled states), reduce sample size
  - How much of Kappa have we gotten?
  - Trade off of Kappa (number of states) vs. runtime